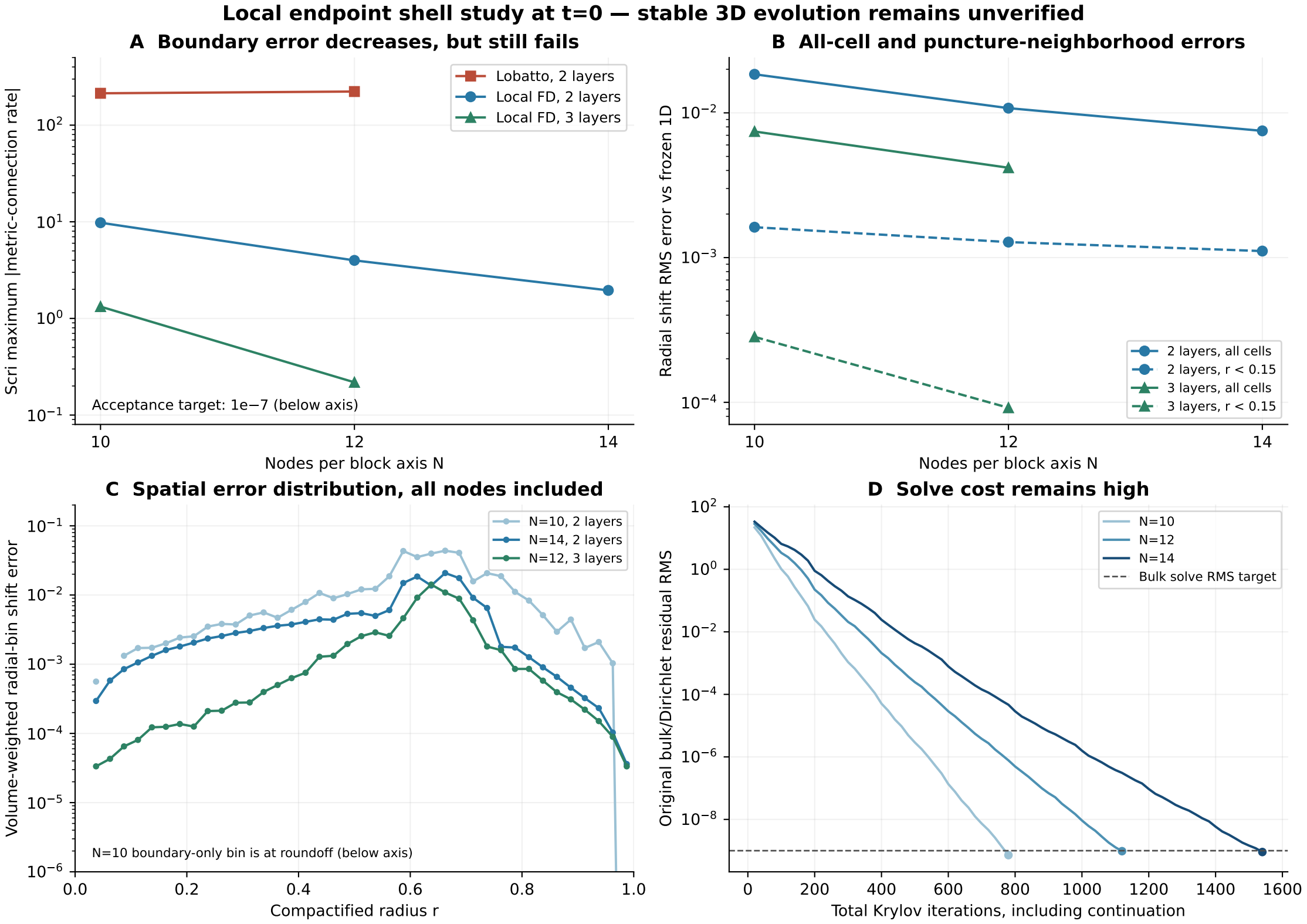
Local endpoint shell study at t=0 — stable 3D evolution remains unverified
A Boundary error decreases, but still fails
Scri maximum |metric-connection rate|
Lobatto, 2 layers
Local FD, 2 layers
Local FD, 3 layers
Acceptance target: 1e−7 (below axis)
Nodes per block axis N
B All-cell and puncture-neighborhood errors
Radial shift RMS error vs frozen 1D
2 layers, all cells
2 layers, r < 0.15
3 layers, all cells
3 layers, r < 0.15
Nodes per block axis N
C Spatial error distribution, all nodes included
Volume-weighted radial-bin shift error
N=10, 2 layers
N=14, 2 layers
N=12, 3 layers
N=10 boundary-only bin is at roundoff (below axis)
Compactified radius r
D Solve cost remains high
Original bulk/Dirichlet residual RMS
N=10
N=12
N=14
Bulk solve RMS target
Total Krylov iterations, including continuation